

LOAN APPROVAL PREDICTION

1. Project Overview

This project focuses on Loan Approval Prediction using historical loan application data. The objective is to analyze applicant information such as income, loan amount, CIBIL score, employment status, education, and asset values to identify the key factors influencing loan approval decisions. The project also aims to build a predictive model that can classify loan applications as Approved or Rejected, helping financial institutions make faster and more accurate lending decisions.

2. Dataset Summary

- Rows: 4,269
- Columns: 13

Key Features

Applicant Information

- Number of Dependents
- Education
- Self Employed Status

Financial Information

- Annual Income
- Loan Amount
- Loan Term
- CIBIL Score

Asset Information

- Residential Asset Value
- Commercial Asset Value

- Luxury Asset Value
- Bank Asset Value

Target Variable

- Loan Status (Approved / Rejected)

Unique Identifier

- Loan ID

Missing Data

- No missing values found in the dataset.

3. Exploratory Data Analysis Using Python

We began with data preparation, cleaning, and analysis using Python.

Data Loading

Imported the dataset using the Pandas library.

Initial Exploration

Used:

```
df.info()
df.describe()
df.head()
```

```
df.info()
```

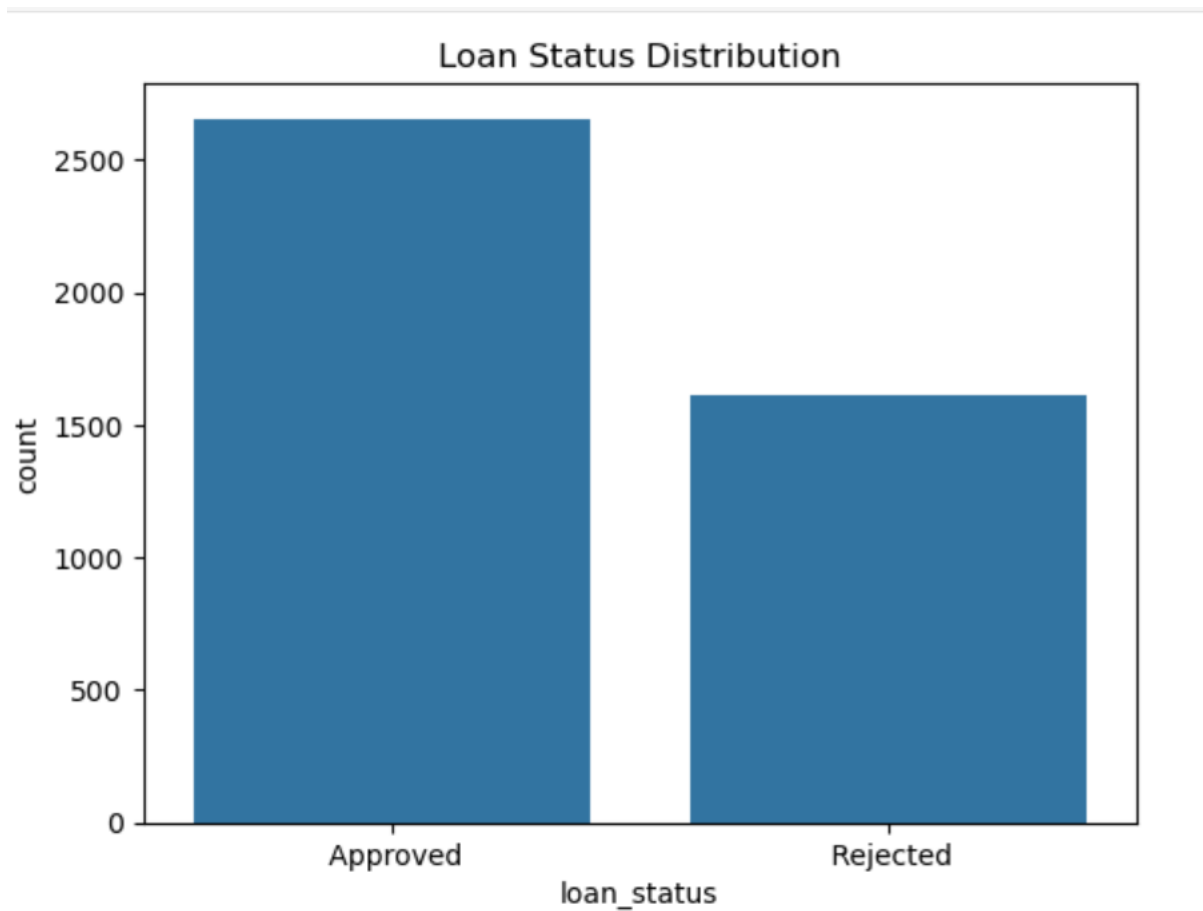
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4269 entries, 0 to 4268
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   loan_id                              4269 non-null   int64
1   no_of_dependents                     4269 non-null   int64
2   education                            4269 non-null   object
3   self_employed                       4269 non-null   object
4   income_annum                         4269 non-null   int64
5   loan_amount                         4269 non-null   int64
6   loan_term                           4269 non-null   int64
7   cibil_score                          4269 non-null   int64
8   residential_assets_value             4269 non-null   int64
9   commercial_assets_value              4269 non-null   int64
10  luxury_assets_value                  4269 non-null   int64
11  bank_asset_value                     4269 non-null   int64
12  loan_status                          4269 non-null   object
dtypes: int64(10), object(3)
memory usage: 433.7+ KB
```

	loan_id	no_of_dependents	income_annum	loan_amount	loan_term	cibil_score	residential_assets_value	com
count	4269.000000	4269.000000	4.269000e+03	4.269000e+03	4269.000000	4269.000000	4.269000e+03	
mean	2135.000000	2.498712	5.059124e+06	1.513345e+07	10.900445	599.936051	7.472617e+06	
std	1232.498479	1.695910	2.806840e+06	9.043363e+06	5.709187	172.430401	6.503637e+06	
min	1.000000	0.000000	2.000000e+05	3.000000e+05	2.000000	300.000000	-1.000000e+05	
25%	1068.000000	1.000000	2.700000e+06	7.700000e+06	6.000000	453.000000	2.200000e+06	
50%	2135.000000	3.000000	5.100000e+06	1.450000e+07	10.000000	600.000000	5.600000e+06	
75%	3202.000000	4.000000	7.500000e+06	2.150000e+07	16.000000	748.000000	1.130000e+07	
max	4269.000000	5.000000	9.900000e+06	3.950000e+07	20.000000	900.000000	2.910000e+07	

Data Visualization

Created visualizations to understand patterns and trends:

- Loan Status Distribution
- Education vs Loan Approval
- Self-Employment Analysis
- Income Distribution
- Loan Amount Distribution
- CIBIL Score Distribution
- Correlation Heatmap



Feature Analysis

Analyzed the impact of:

- Annual Income
- Loan Amount
- CIBIL Score
- Asset Values
- Education
- Self Employment Status

on loan approval decisions.

Machine Learning Model

Built a **Random Forest Classifier** model to predict loan approval status.

Model Evaluation

Evaluated model performance using:

- Accuracy Score
- Confusion Matrix
- Classification Report

Accuracy: 0.9742388758782201

```
[[527   9]
 [ 13 305]]
```

	precision	recall	f1-score	support
0	0.98	0.98	0.98	536
1	0.97	0.96	0.97	318
accuracy			0.97	854
macro avg	0.97	0.97	0.97	854
weighted avg	0.97	0.97	0.97	854

4. Data Analysis Using SQL (Loan Approval Analysis)

We performed structured analysis in SQL to answer key business and financial questions related to loan approval decisions.

Key SQL Analysis Performed

1. Total Loan Applications

Calculated the total number of loan applications received by the financial institution.

	total_applications bigint
1	4269

2. Average Income

This query calculates the average annual income of all applicants.

	average_income numeric
1	5059123.916608104943

3. Average Loan Amount

This query calculates the average loan amount requested by applicants.

	average_loan_amount numeric
1	15133450.456781447646

4. Education-Wise Loan Applications

This query groups applicants by education level and counts the number of applications in each category.

	education bigint	total_applications bigint
1	0	2144
2	1	2125

5. Self-Employed vs Loan Status

This query compares loan approval and rejection counts for self-employed and non-self-employed applicants.

	self_employed bigint	loan_status bigint	total bigint
1	1	1	812
2	0	0	1318
3	1	0	1338
4	0	1	801

6. Average CIBIL Score by Loan Status

This query calculates the average CIBIL score for approved and rejected applicants.

	loan_status bigint	avg_cibil numeric
1	0	703.4619728915662651
2	1	429.4680719156850589

7. Top 10 Applicants with Highest Income

This query retrieves the top 10 applicants with the highest annual income.

	loan_id bigint	income_annum bigint
1	531	9900000
2	1028	9900000
3	361	9900000
4	510	9900000
5	715	9900000
6	782	9900000
7	158	9900000
8	301	9900000
9	83	9900000
10	1132	9900000

8. Highest Loan Amount

This query finds the maximum loan amount requested in the dataset.

	highest_loan bigint
1	39500000

9. Average Loan Amount by Education

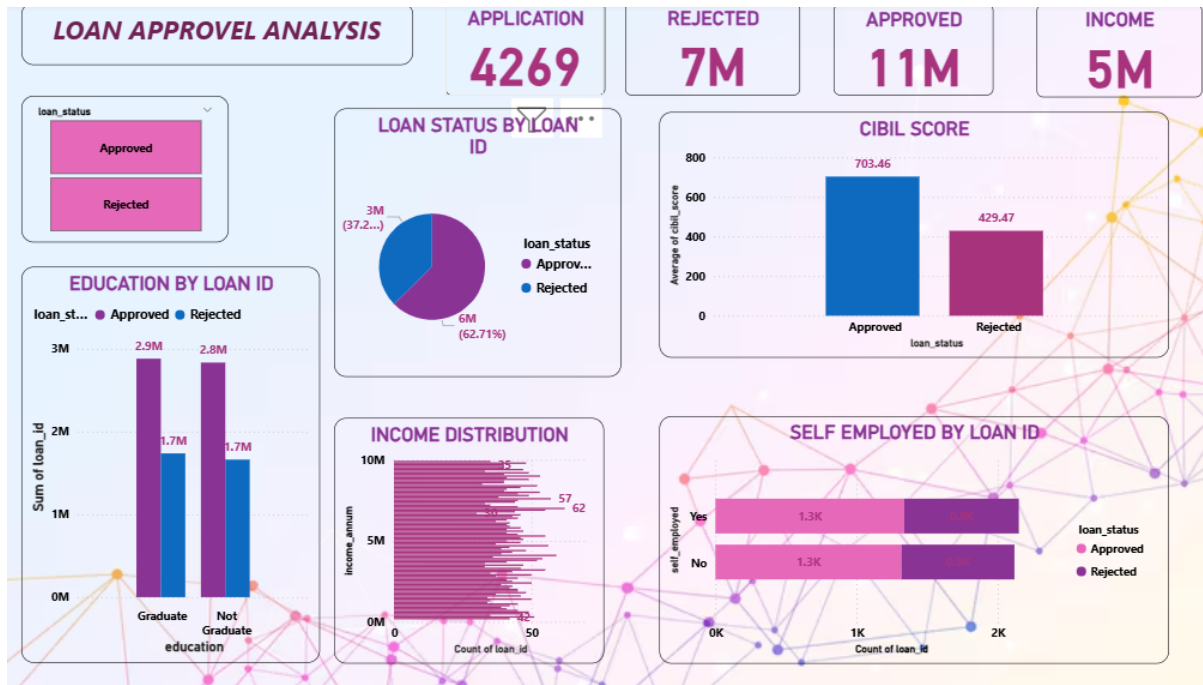
	education bigint	avg_loan_amount numeric
1	0	15229151.119402985075
2	1	15036894.117647058824

10. Income Category Analysis

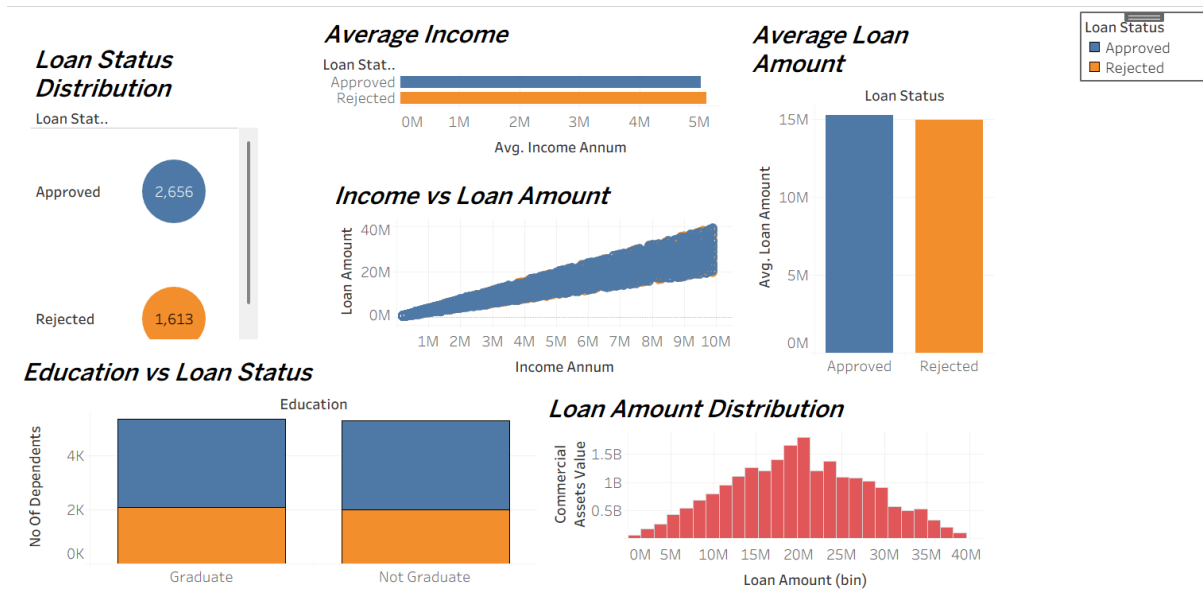
This query classifies applicants into Low, Medium, and High Income groups and counts the number of applicants in each category.

	income_category text	total_applicants bigint
1	Low Income	1199
2	High Income	1249
3	Medium Income	1821

5. Dashboard in power bi



6. Dashboard in tableau



7. Business Recommendations

Based on the Loan Approval Prediction analysis, the following recommendations can help financial institutions improve loan approval processes and reduce risk:

1. Prioritize Applicants with High CIBIL Scores

- Applicants with higher CIBIL scores are more likely to repay loans on time.
- Give preference to applicants with strong credit histories.
- Implement minimum CIBIL score requirements for specific loan categories.

2. Improve Risk Assessment for Low CIBIL Applicants

- Conduct additional verification for applicants with low credit scores.
- Request collateral or guarantors before approval.
- Offer smaller loan amounts to high-risk customers.

3. Consider Income-Based Loan Eligibility

- Higher-income applicants generally have better repayment capacity.
- Establish loan amount limits based on annual income.
- Use debt-to-income ratios to evaluate affordability.

4. Leverage Asset Information

- Applicants with higher residential, commercial, and bank asset values represent lower lending risk.
- Incorporate asset valuation into loan approval criteria.
- Provide favorable terms to applicants with strong asset portfolios.

5. Develop Customized Loan Products

- Create different loan plans for:
 - Salaried employees
 - Self-employed individuals
 - Small business owners
- Tailor interest rates and repayment terms based on customer profiles.

6. Use Automated Loan Approval Systems

- Implement machine learning models to assist in approval decisions.
- Reduce manual processing time.
- Improve consistency and accuracy in loan evaluation.